

MODEL QUESTION PAPER

REVISION 2010
SUB CODE:3039

Reg.No:.....
Signature.....

THIRD SEMESTER DIPLOMA EXAMINATION IN MECHANICAL ENGINEERING

METTALLURGY AND MACHINE TOOLS

Time : 3 hrs

Marks : 100

PART A

I. Answer all questions in one or two sentences.

1. Define crystal defects.
2. Name the furnace in which the pig iron is manufacturing.
3. Express Gibb's phase rule equation and write the notations used in it.
4. Write the tool signature of a single point cutting tool.
5. Name two quick return mechanisms used in shaper machine.

5X2=10

PART B

II. Answer any five questions from the following.

1. Differentiate slip and twinning with neat sketch.
2. Draw the cooling curve of pure iron and show the critical points.
3. Define cutting speed, feed and depth of cut.
4. Mention the desirable properties of cutting fluids.
5. With a neat sketch explain the crank and slotted lever mechanism of shaper.
6. Compare the shaper and slotter
7. List out the specifications a milling machine.

5X6=30

PART C MODULE I

III.

- a. Illustrate the crystal growth and grain formation in metallic crystals. 7
- b. Explain the types of cast iron with sketch of crystal structure. 8

OR

IV

- a. With neat sketches explain the line defects in crystal structure. 7
- b. Explain L-D process with neat sketch. 8

MODULE II

V

- a. Classify and explain the crystal structures of alloys. 7
- b. Draw the Iron – Carbon equilibrium diagram and show the various phases and critical lines. 8

OR

VI

- a. Explain the TTT diagram. 7
- b. Explain the following
 - i. Annealing ii. Normalizing iii. Hardening iv. Tempering. 8

MODULE III

VII.

- a. List the properties required for a good Lubricant. 7
- b. Describe any two taper turning methods with sketch. 8

OR

VIII.

- a. Show the cutting forces acting on a single point cutting tool and explain 7
- b. Explain any four drilling machine operations. 8

MODULE IV

IX.

- a. Draw the line sketch of slotter machine and state the working principle. 7
- b. List the work holding devices using in milling machine. 8

OR

X.

- a. Explain the automatic feed mechanism in shaper with a neat sketch. 7
- b. List the various milling operations and explain any two. 8
